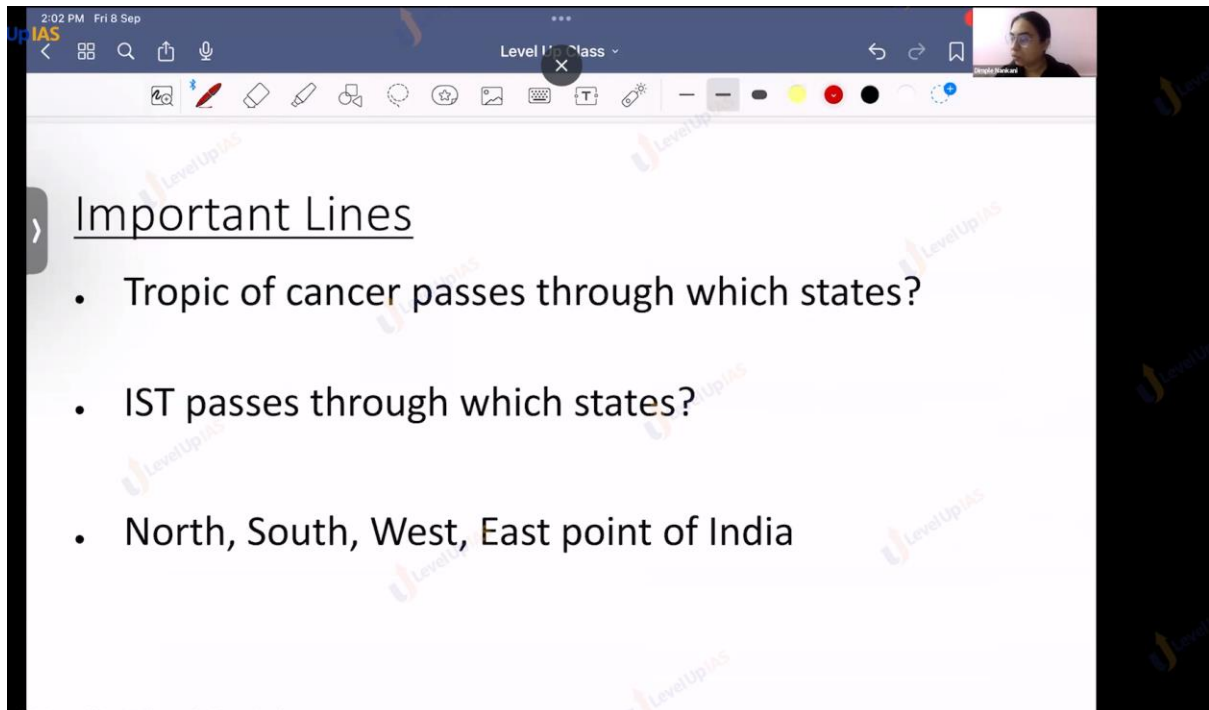


(30) North plane and rivers



The screenshot shows a video lecture interface. At the top, there is a header bar with the text 'Level Up IAS' and a dropdown menu 'Level Up IAS'. Below the header, there is a toolbar with various icons for drawing and editing. The main content area displays the title 'Important Lines' followed by a list of three questions:

- Tropic of cancer passes through which states?
- IST passes through which states?
- North, South, West, East point of India

Mapping India:

-
1. Tides refers to vertical movement of water nearly each place on the earth experiences two high tide tow low tide
-

5 major movements - PP

- ↳ Rifting → coal - N, S, G, D, M
- ↳ Formation of DLP → French Reunion
 - ↳ Deccan lava Trap. - GJ, MH, KAN
- ↳ Formation of Waghat
- ↳ Marine Transgression
- ↳ Malda Fault

Five major movements in the peninsular plate.

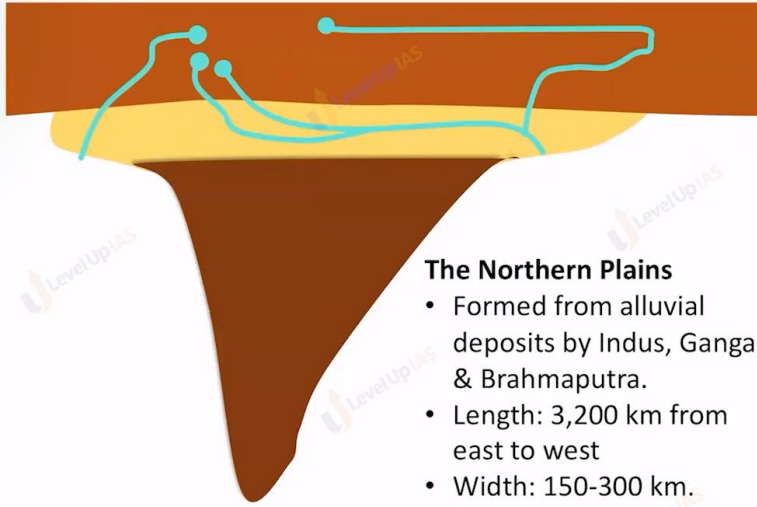
2:12 PM Fri 8 Sep
Level Up Class

Details of Northern Mountain Complex

- Series of ranges:
- **Trans Him:** Karakorum, Ladakh, Zaskar: Formed by Upliftment of Tethys Sea Sediments: O-C Collision
- **Great Him:** Continuous Range: C-C Collision: Formed by Upliftment of marine sediments: sedimentary and metamorphic
- **Middle Himalayas:** C-C Collision: Formed by Upliftment of marine sediments: sedimentary and metamorphic
 - Broken Ranges and are locally known by different names like:
 - Pir Panjal in Jnk and H.P
 - Dhauladhar in H.P
 - Mussorie, Nagtibba, Garhwal in UTK
- **Shivalik: C-C Collision:** Formed by upliftment of fluvial deposits
- **Purvanchal Range:** CC Collision: Marine Sediments.

delve

Formation of northern plains



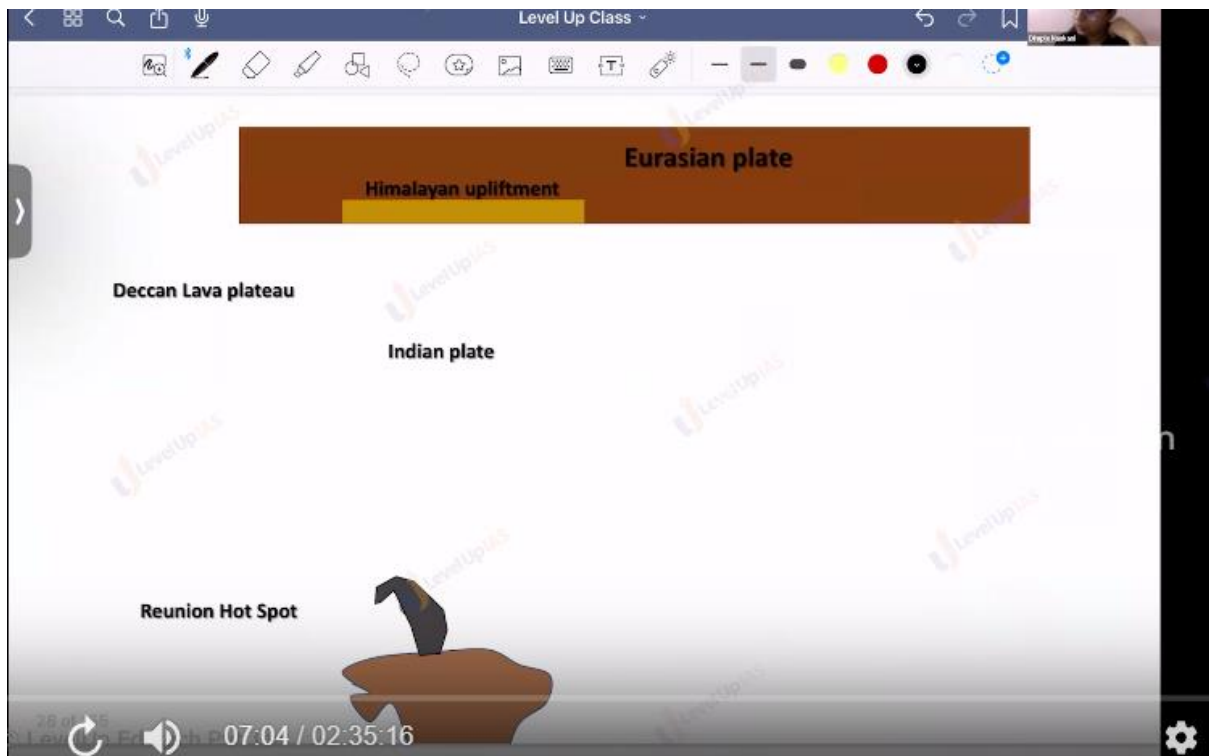
The Northern Plains

- Formed from alluvial deposits by Indus, Ganga & Brahmaputra.
- Length: 3,200 km from east to west
- Width: 150-300 km.
- Depth: 1,000-2,000 m.

Rifting

1. Due to Plate movement, PP has undergone **vertical movements** like **faulting, rifting**.
2. **Eg:** Rift valleys of Narmada, **Tapi**, **Mahanadi**, **Son**, **Damodar**, **Subarnarekha**. **Rivers flow via these rift valleys** and these are associated with the **coal fields**.
3. River valleys in peninsular India are **shallow with low gradients**.





North Eastern Plateau: Meghalayan Plateau

- Meghalaya plateau is further sub-divided into three:
 - (i) The Garo Hills;
 - (ii) The Khasi Hills;
 - (iii) The Jaintia Hills
- Extension of this is also seen in Karbi Anglong hills of Assam.
- Similar to the Chotanagpur plateau, the Meghalaya plateau is also rich in mineral resources like coal, iron ore, sillimanite, limestone and uranium.
- This area receives maximum rainfall from the south west monsoon. As a result, the Meghalaya plateau has a highly eroded surface. Cherrapunji displays a bare rocky surface devoid of any permanent vegetation cover.

ABODE OF CLOUDS
The East Khasi Hills in Meghalaya get a chunk of rain

11,873 mm
average yearly rainfall

MEGHALAYA - SHILLONG

MAWSYNRAM

BANGLADESH

2:08 PM Fri 8 Sep
Level Up Class

Marine Transgression

- As India Plate moved northward, there are 4 events of marine transgression (sea water entering into Peninsular Plateau)
- These areas form the major oil and gas resources
 - Kutch Saurashtra Shelf: South Bassein, Bombay High
 - K G Basin: Rawa Oil Fields
 - Assam Region: Naharkatiya, Sibsagar, Moran Hugrijan.

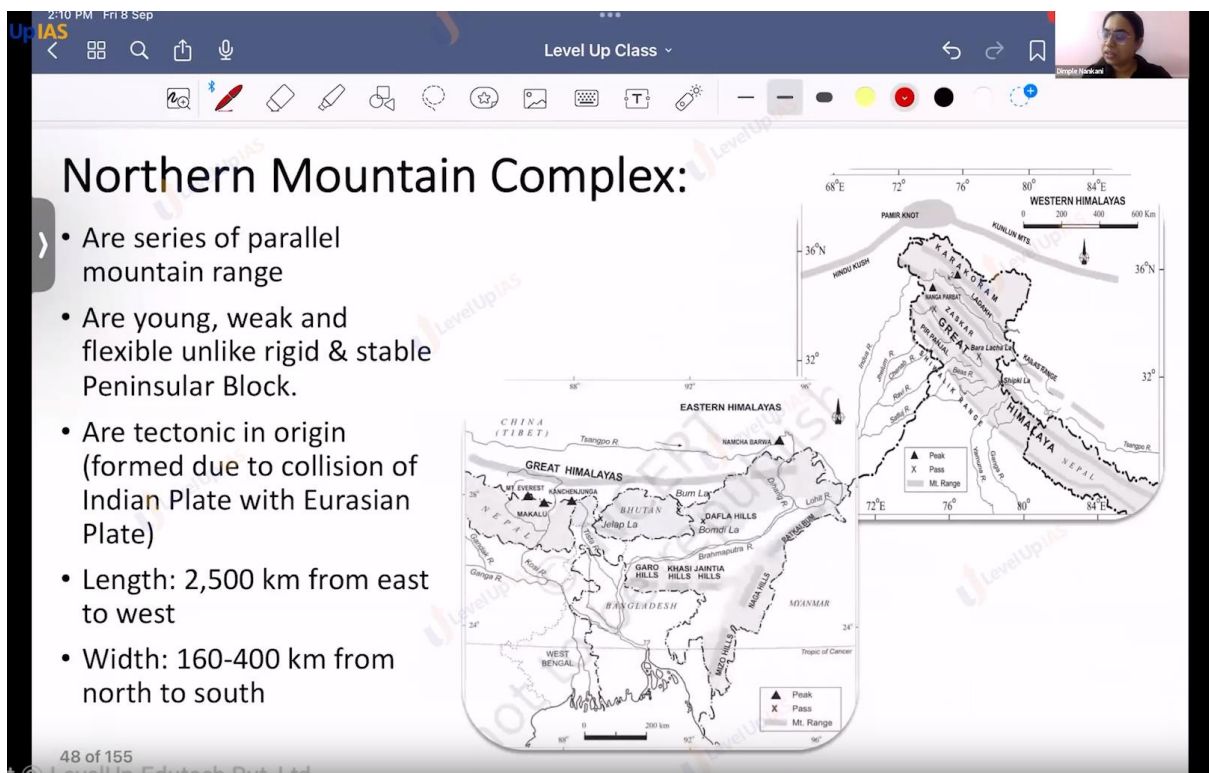
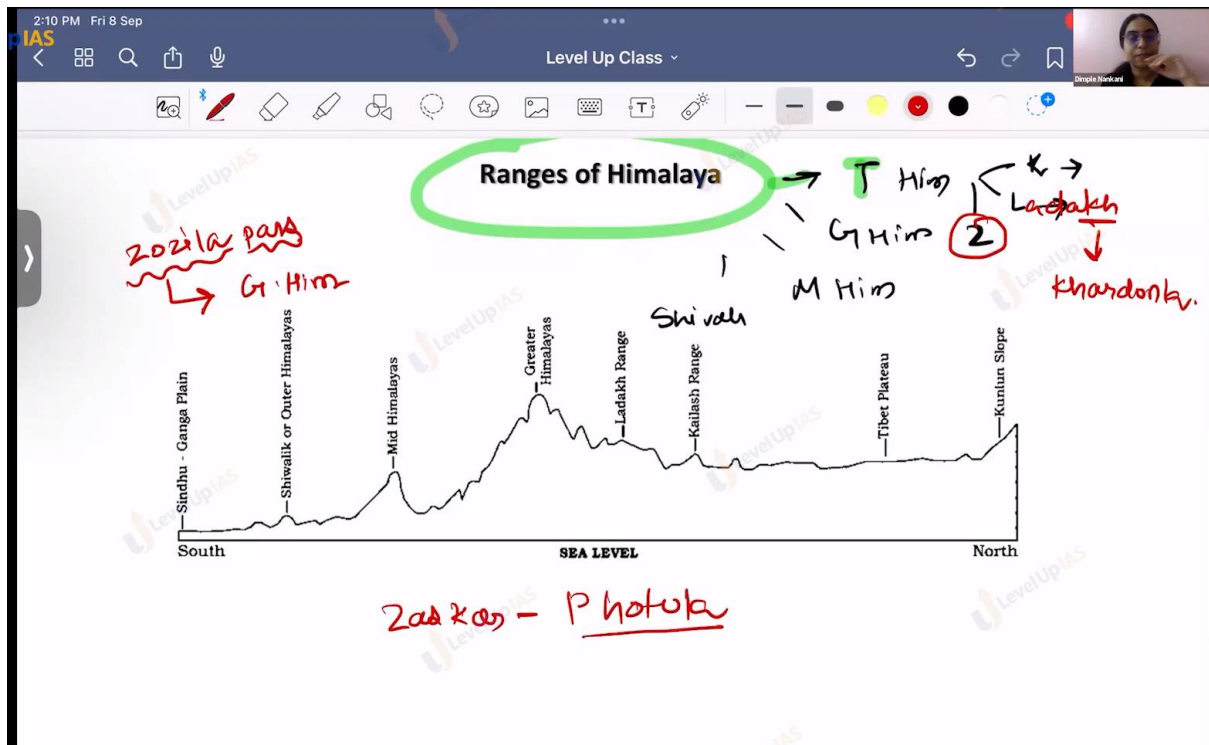
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Level Up Class

Deccan Plateau: Western Ghat

- W. Ghats** has higher elevation and are more **continuous** than E. Ghats.
- Western Ghats: Elevation is 1,500 m with the height **increasing from north to south**.
- Anaimudi**= Highest peak of Peninsular plateau on Anaimalai hills
- Dodabetta** is highest peak of Nilgiri hills.
- Peninsular rivers** have their **origin** in Western Ghats.

07:45 / 02:35:16



2:10 PM Fri 8 Sep
Level Up Class

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Level Up Class

Door Slam Shut

Sequence of collision of Peninsular Plateau

- Collision first happened on Western Side
- Plateau then rotated in anti clockwise manner and got hinged on the NE side and western side got opened.
- Opening of Western Edge led to slumping of sediments on W. Him whereas Eastern Him continues to be squashed.
- This explains why Western Himalayas are broader while the Eastern Himalayas do not have distinct range.



Details of Northern Mountain Complex

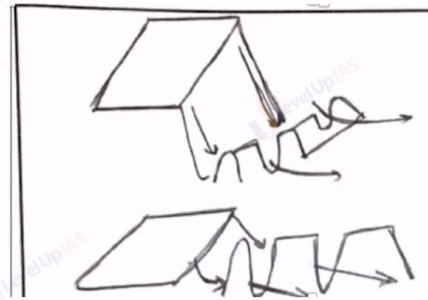
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Eastern Himalayas V/s Western Himalayas

Eastern Himalayas <i>Western</i>	Western Himalayas <i>Eastern</i>
Broader: 400 km: N-S ✓	Narrower: 250 km: N-S ✓
Ranges are <u>more distinct</u> : TH, GH, MH, Shivalik.	Ranges are not distinct: TH, GH, MH, Shivalik cannot be distinguished.
They are not as steep. They rise up as a series of steps	Very Steep. Rise abruptly from foothills
They have more Hill station and Glaciers	They have less Hill station and Glaciers
More Drier	More Wetter, Moist
Coniferous, Deciduous forest	Evergreen Forest

Himalayas are not water divide

1. Almost all the Himalayan rivers cut across mountains and drain into N. Plain
2. Such Rivers are called as **Antecedent Rivers**. Rivers are older than Himalayas as Himalayas uplifted the rivers could cut across the Himalayas.
3. This happened as rivers were large enough and Himalayas were rising at the rate, were the rivers could keep pace and erode
4. When **Mountains were being uplifted**, they rivers form Tibetan highlands were blocked and formed longitudinal lakes between the ranges.
5. Eventually as the rivers cut across the lakes were drained leaving behind the dry lake beds called as **DUN/WADI between Him Ranges**
6. These are location for **agriculture and Horticulture** cause of fertile lacustrine Deposits



Longitudinal valley is an elongated valley found between two almost-parallel mountain chains in geologically young fold mountains

Lacustrine Means Derived from Lake

2:14 PM Fri 8 Sep Level Up Class

Division of Northern mountains

Kashmir or Northwestern Himalayas

➡

Himachal and Uttarakhand Himalayas

➡

Darjeeling and Sikkim Himalayas

➡

Arunachal/Assam Himalayas

➡

Eastern Hills and Mountains

REGIONAL DIVISIONS OF THE HIMALAYAS

Punjab Himalayas:

- This part lies between the Indus and Sutlej. From west to east, this is also known as Kashmir Himalaya and Himachal Himalaya; respectively.

Kumaon Himalayas:

- This part lies between Sutlej and Kali rivers.

Nepal Himalayas:

- This part lies between the Kali and Tista rivers.

Assam Himalayas:

- This part lies between the Tista and Dihang

Eastern hills

- Patkai Hills, Naga Hills, Manipuri Hills and Mizo Hills.

55 of 155

India Map Location

2013:

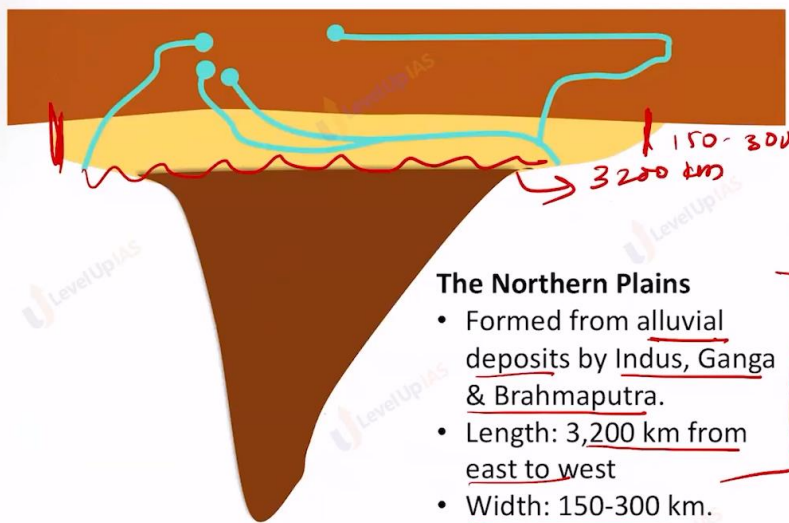
Consider the following pairs:

1. Nokrek Biosphere Reserve : Garo Hills
2. Loktak Lake : Barail Range
3. Namdapha National Park : Daffa Hills

Which of the above pairs is/are correctly matched?

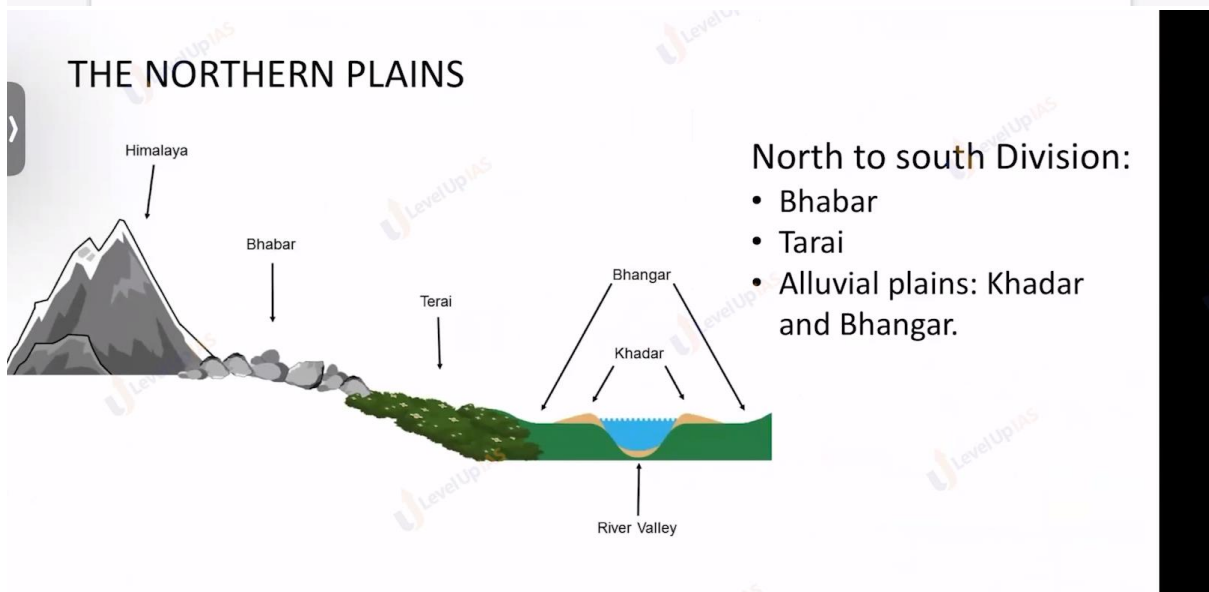
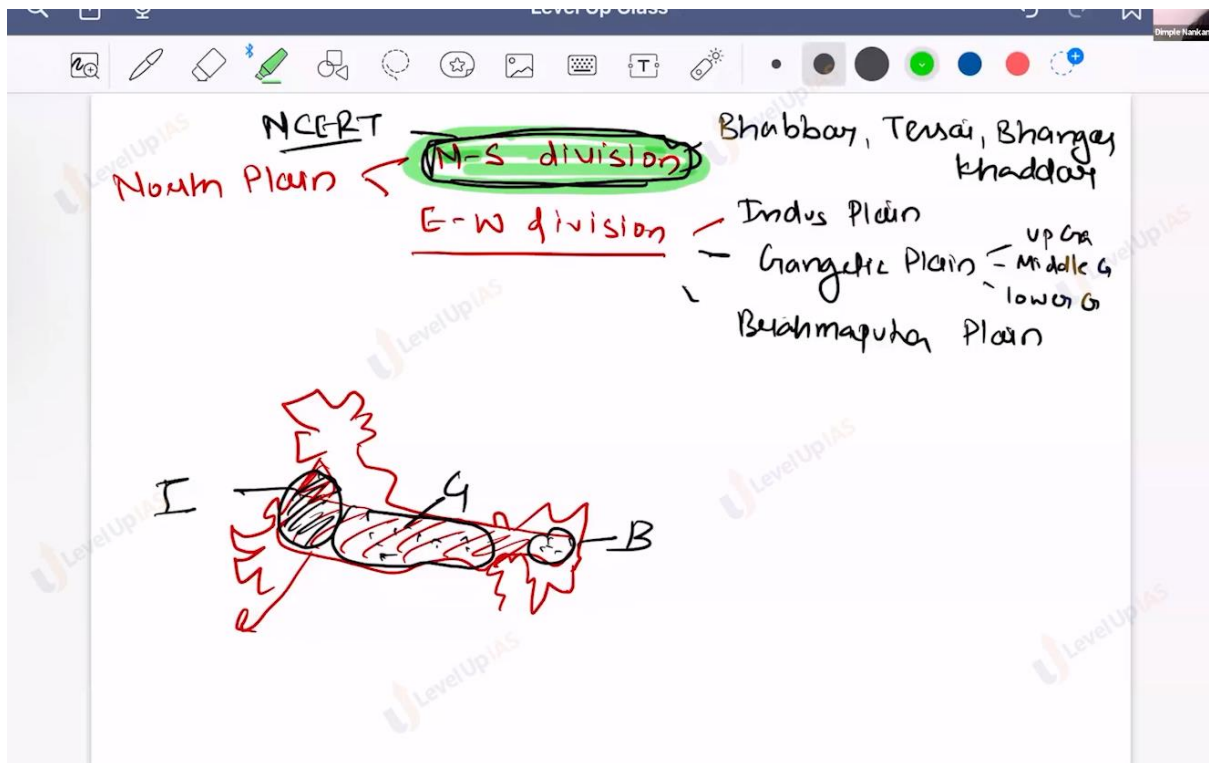
- (a) 1 only
(b) 2 and 3 only
(c) 1, 2 and 3
(d) None

Formation of northern plains

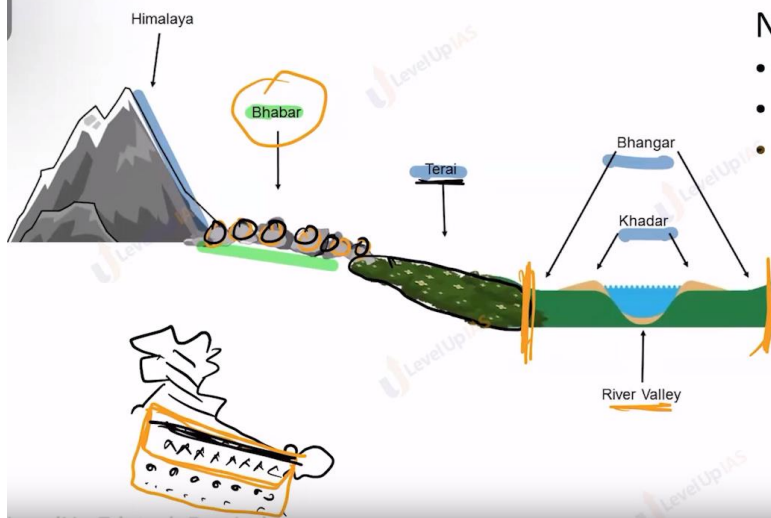


The Northern Plains

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- Length: 3,200 km from east to west
- Width: 150-300 km.
- Depth: 1,000-2,000 m. 1 km-2 km



THE NORTHERN PLAINS



North to south Division:

- Bhabar
- Terai
- Alluvial plains: Khadar and Bhangar.

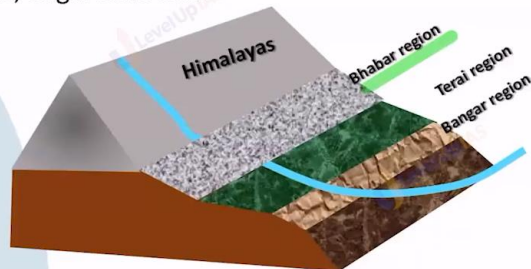
PLAINS OF THE NORTHERN INDIA

THE BHABAR PLAINS

- Narrow belt: 8-10 km parallel to Shiwalik
- Lie to the south of Shiwalik
- Jammu to Assam.
- Rivers coming from mountains deposit bigger rocks, boulders
- River disappear in this zone
- Not suitable for cultivation: Forested Areas and Big trees

THE TARAI PLAIN

- South of the Bhabar tract
- Width: 10-20 km
- Rivers re-emerge here creating marshy-swampy conditions
- Good for the cultivation of Wheat, rice, maize, sugarcane etc.



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LevelUp IAS
LevelUp Class

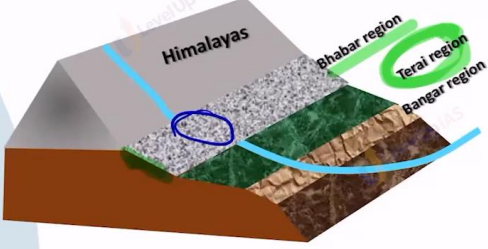
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Handwritten notes: 'agwix' with an arrow pointing to 'River disappear in this zone'.

Alluvial Plain: Bhangar and Khadar

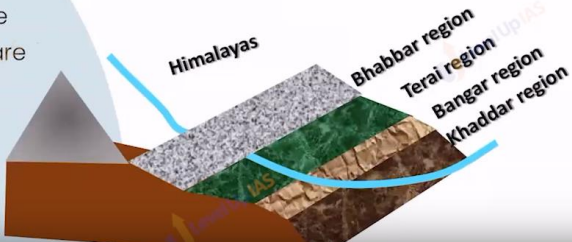
- Features: Sand bars, meanders, oxbow lakes and braided channels.
- Have periodic floods and shifting river courses
- Mouths of rivers has largest deltas of the world. For eg: Sunderbans delta

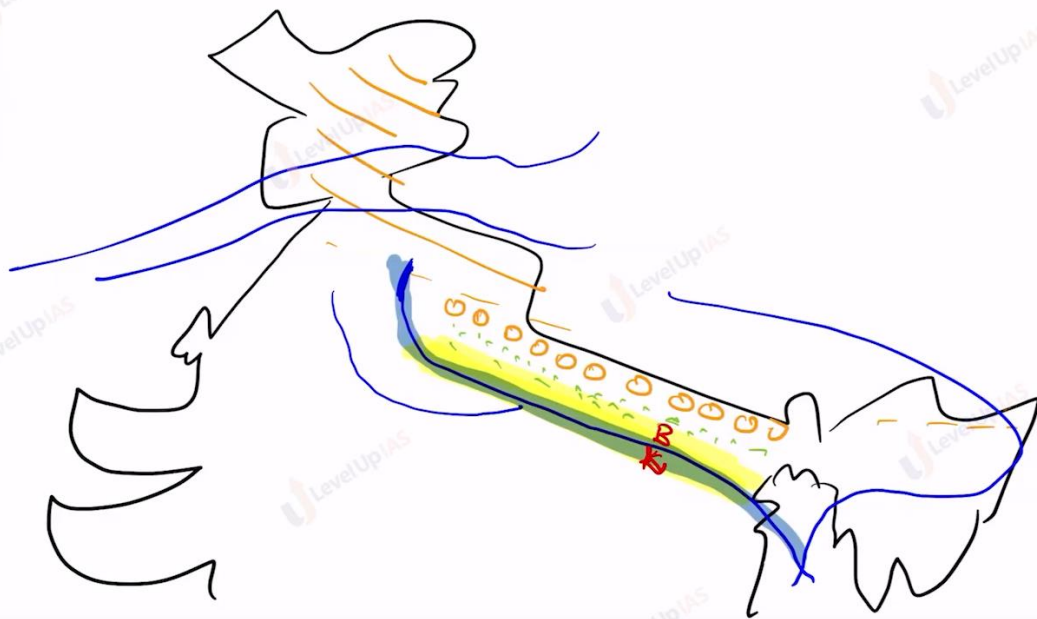
THE BHANGAR PLAINS

- Older alluvial plains.
- Contains the calcium carbonate nodules called 'Kankars' which are impure in nature.

THE KHADAR PLAINS

- Newer Alluvial Plain
- Enriched by fresh deposits of silt every year.
- Consists of silt, mud, clay, and sand.
- Cultivation of sugarcane, rice, wheat, maize and oilseeds.





Alluvial Plain: Bhangar and Khadar

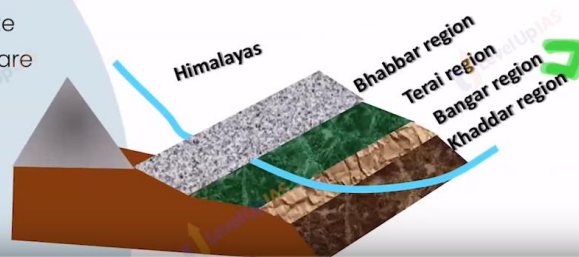
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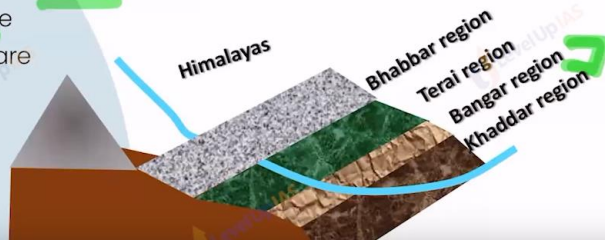
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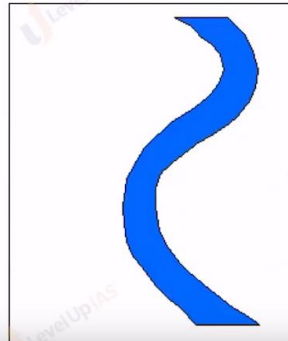
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Features of North Plain

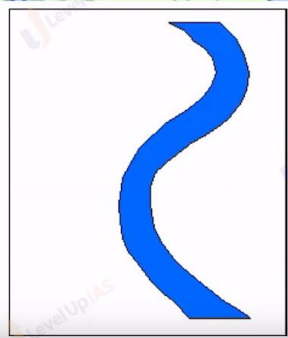
1. Form some of the largest deltas of the world like Sunderbans delta.
2. Generally featureless plain with an elevation of 50-150 m
3. Haryana and Delhi form a water divide between the Indus and the Ganga river systems.
4. Brahmaputra river flows from the northeast to the southwest direction before it takes an almost 90° southward turn at Dhubri before it enters into Bangladesh.
5. River valley plains have a fertile alluvial soil cover which supports a variety of crops like wheat, rice, sugarcane and jute, and hence, supports a large



2:38 PM Fri 8 Sep
Level Up Class

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- H
- NP
- PP

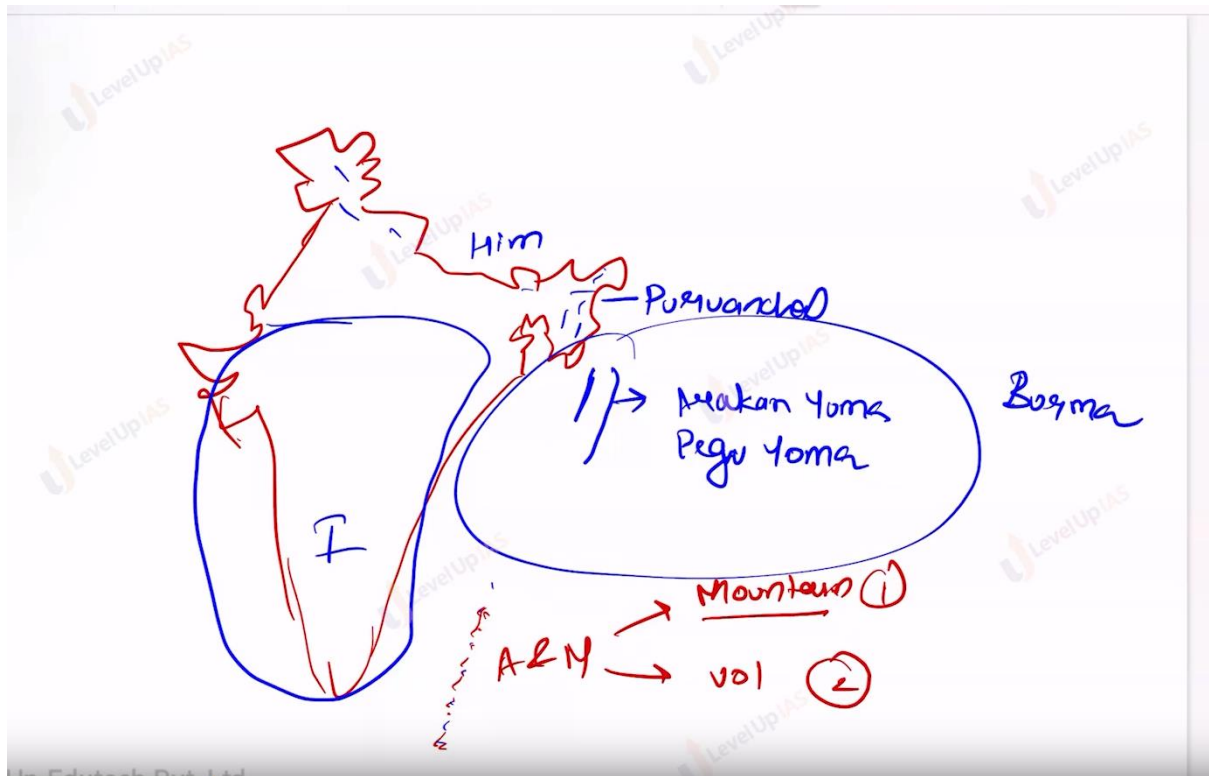
Islands: 2 Major Group

- Andaman and Nicobar:
 - 572 islands/islets.
 - Between 6°N-14°N and 92°E -94°E.
 - The two principal groups of islets include the Ritchie's archipelago and the Labrynth island.
 - Two broad Cateogory: Andaman in the north and the Nicobar in the south separated by a waterbody which is called the Ten degree channel.
 - Some believe it is an elevated portion of submarine mountains.
 - However, some smaller islands are volcanic in origin.
 - Barren island, the only active volcano in India is also situated in the Nicobar islands
 - Has corals and has equatorial vegetation

Islands: 2 Major Group

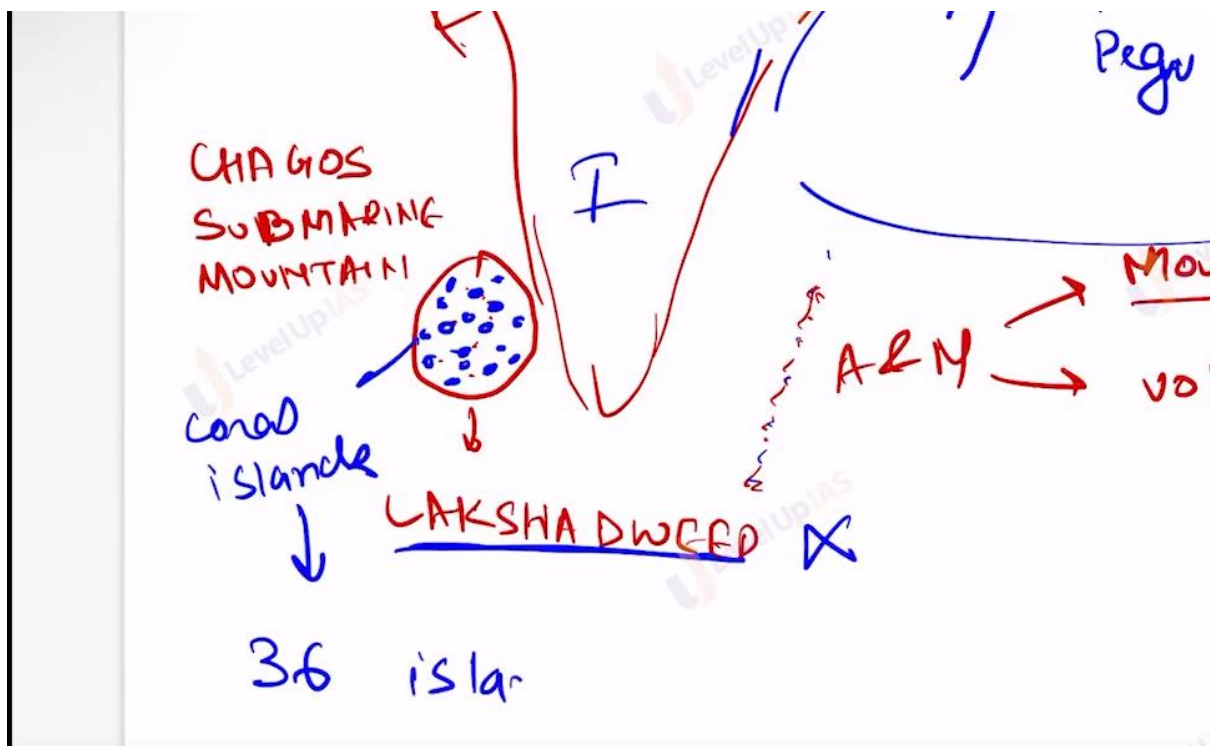
- Lakshadweep:

- 8°N - 12°N and 71°E - 74°E longitude.
- Entire island is coral deposits.
- 36 islands of which 11 are inhabited.
- Minicoy is the largest island
- Islands are divided by the Ten degree channel, north of which is the Amini Island and to the south of the Canannore Island..



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2021

With references to India, Didwana, Kuchaman, Sargol and khatu are the names of

- (a) Glaciers
- (b) Mangrove areas
- (c) Ramsar sites
- (d) Saline lakes

2021

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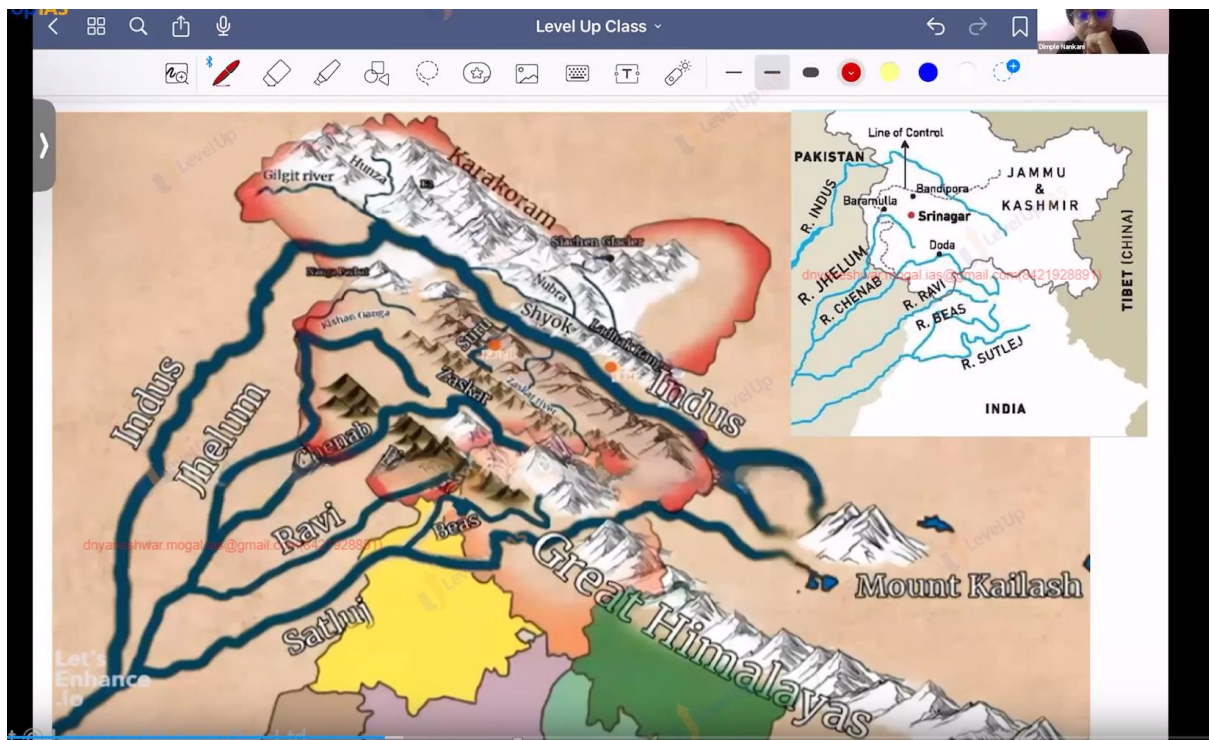
2018

Consider the following statements:

1. The Barren Island volcano is an active volcano located in the Indian territory.
2. Barren Island lies about 140 km east of Great Nicobar.
3. The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.

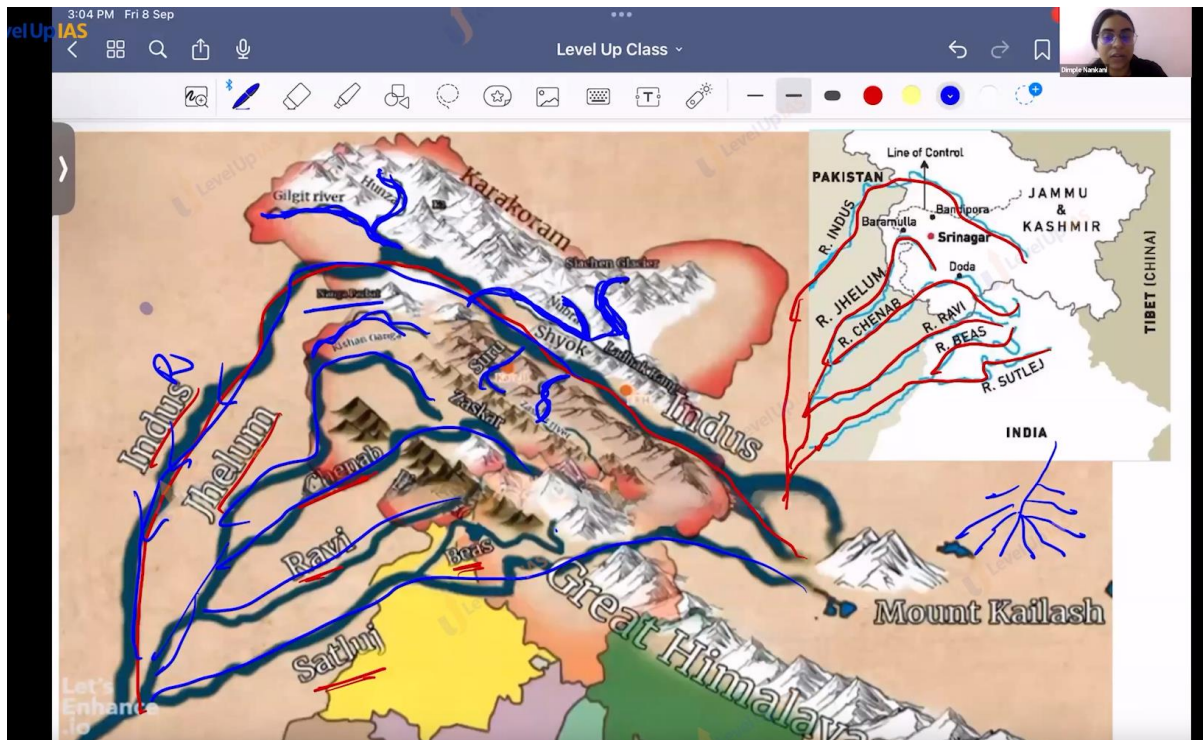
Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3
- (c) 3 only
- (d) 1 and 3



1 - River → source
 → tributary
 → special features → dams - State
 → National Park - State
 → pilgrim centre - State

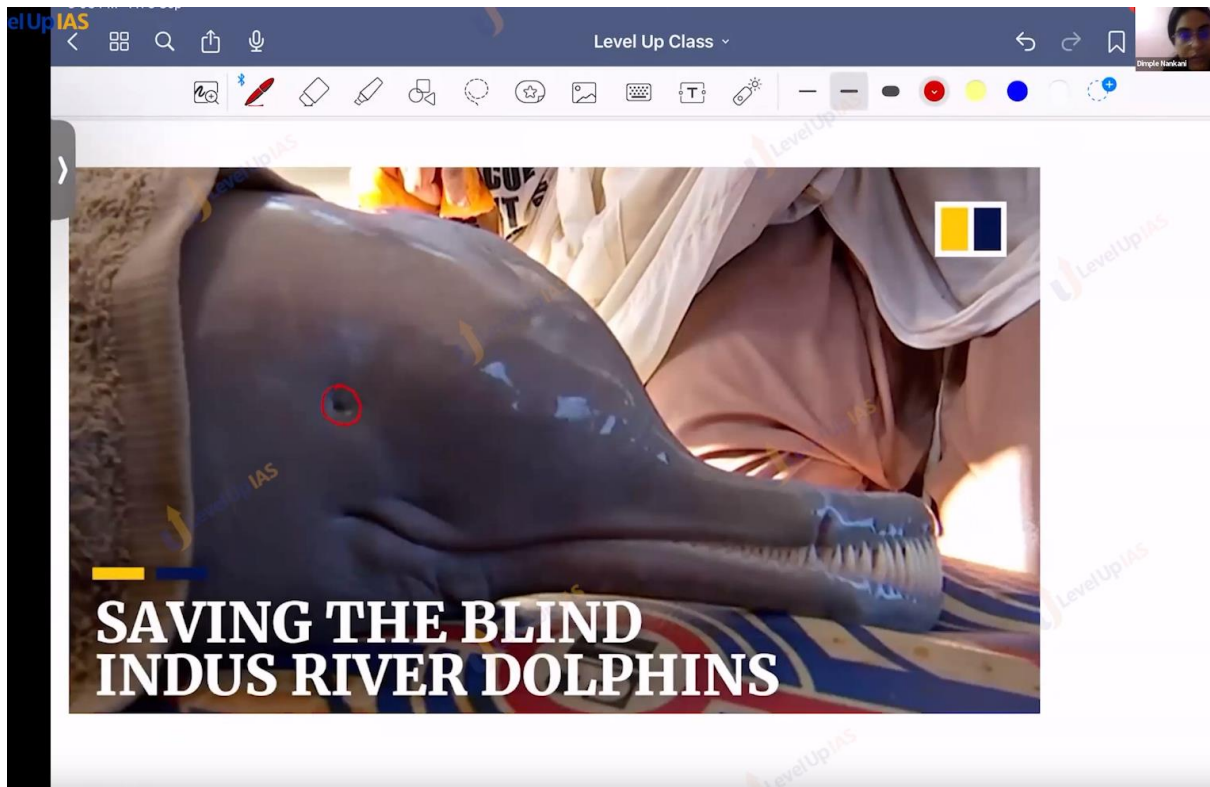
- Indus - trib - Ganges - trib - Beahmaputra - Yamuna & tributaries - Teesta	- Godavari - Krishna - Mahanadi - Kaveri - Narmada - Tapi - Eastern Rivers - Bareilly	- Damodar - Mahanadi - Sabarnnathi - Beahmani - Baiterhi - Sabarnathi - Chambas - Yamuna
--	--	---



Level Up Class

Indus:

- Origin: Bokarchu Glacier (West of Mansarovar)
- Has Blind Indus Dolphin
- Aka Singhi Khambab in Tibet
- Lake: Satpara Lake
- Hemis High Altitude National: Largest and Highest NP, Snow Leopard
- Imp left Bank Tributary: Suru, Zaskar River
- River Shyok, Gilgit (Tributary: Hunza) are right bank tributary.
- River Shyok has origin in Siachen



INDUS RIVER	
LEFT TRIB.	RIGHT TRIB.
Jhelum	Shyok
Chenab	Hunza, Gilgit
Ravi, Beas	Swat
Satluj	Kunar, Kabul
Zaskar, Panjnad	Kurram

Level Up Class

Jhelum:

- Origin: Verinag Spring at foot of PirPanjal
- Tributary: Neelam, Kishenganga
- Has wular lake, Dal Lake, Srinagar
- Forms border between India and Pakistan
- Meets Chenab in Pakistan

Revising the indus.

Indus

origin - Bokaranogh

R.B - Gilgit, Shyok

↓ ↓

Gilgit Shachen

R.B - Suro, Zaskar

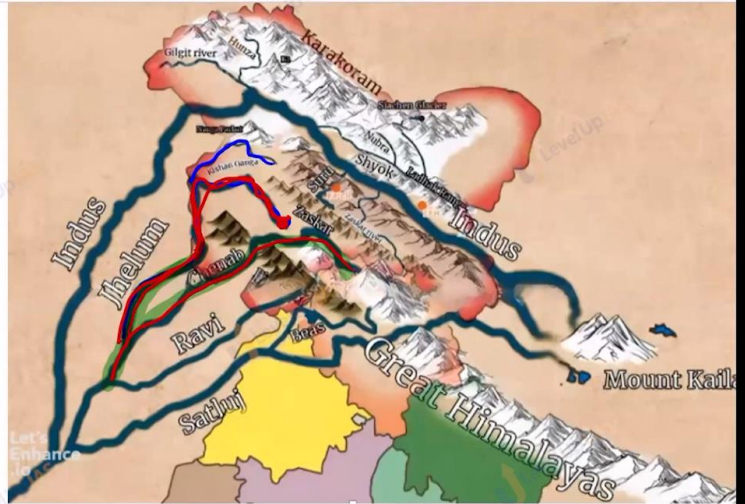
- J, C, R.B.S

NP - Hemis Hig' → SN ow 2

Fama - B ≠ P

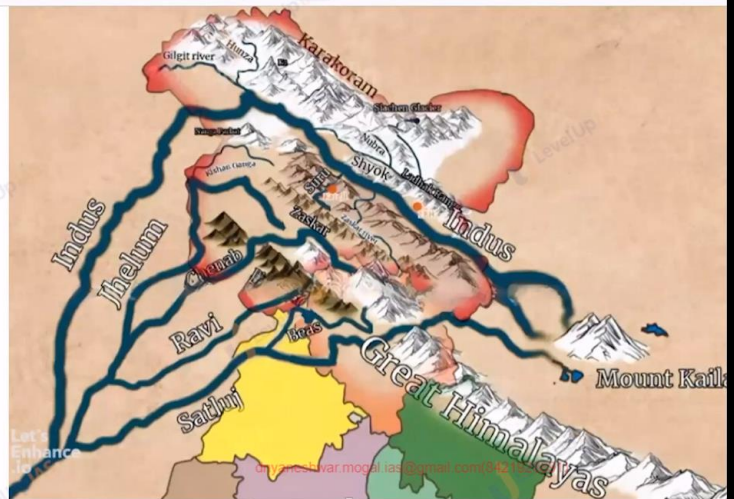
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Chenab:

- Made of Chandra and Bhaga: Both meets at Keylong: Hence called as Lady of Keylong
- Origin: Baralacha La Pass, Barashigri Glacier
- Largest Tributary of Indus
- Meets Indus River



Level Up Class

Ravi

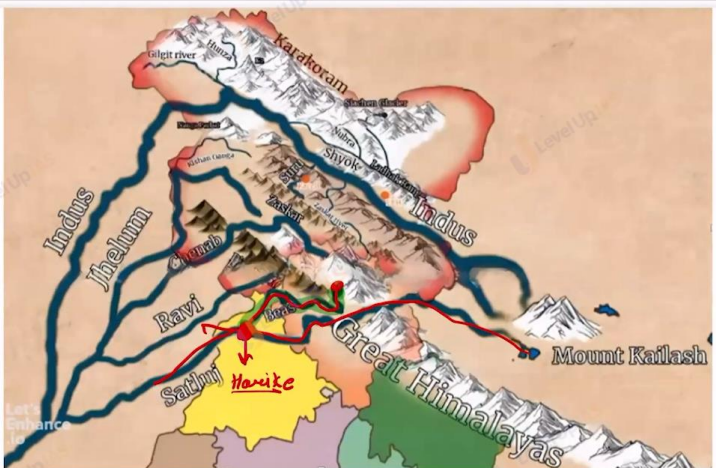
- Origin: Rohtang Pass (HP) in Kullu
- Between Pir Panjal and Dhauladhar
- Cuts Dhauladhar
- Dalhousie is on Ravi
- Meets Chenab River in Pakistan



Level Up Class

Beas:

- Origin: Beas Kund (HP) near Rohtang Pass
- Dam: Pong Dam aka Maharana Pratap Sagar Dam
- Has Pong Wetland, Harike Wetland
- Flows entirely in India
- Beas and Sutlej meet at Harike to form Harike wetland



LevelUp IAS

Level Up Class

Sutlej:

- Origin Rakas Lake near Mansarovar
- Aka Langchen Khambab in Tibet
- Passes through Shipki la Pass in Dhauladhar Range
- Dam: Bhakra Nangal Dam aka Govind Sagar Dam (HP);
- Has Harike Wetland
- Meets River Chenab in Pakistan

2021: UPSC

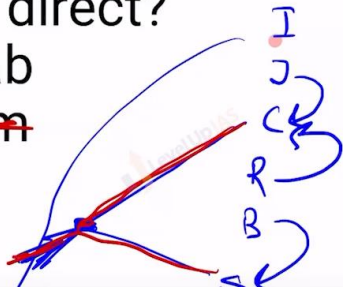
With reference to the Indus River system, of the following four rivers, three of them pour into one of the which joins the Indus direct. Among the following, which one is such river that joins the Indus direct?

- (a) Chenab
- (b) Jhelum
- (c) Ravi
- (d) Sutlej

2021: UPSC

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- (a) Chenab
- (b) ~~Jhelum~~
- (c) ~~Ravi~~
- (d) ~~Sutlej~~



2020: UPSC

Siachen Glacier is situated to the

- (a) ^{west} ~~East~~ of Aksai Chin
- (b) ^{northw} ~~East~~ of Leh
- (c) ^{East} ~~North~~ of Gilgit
- (d) North of Nubra Valley

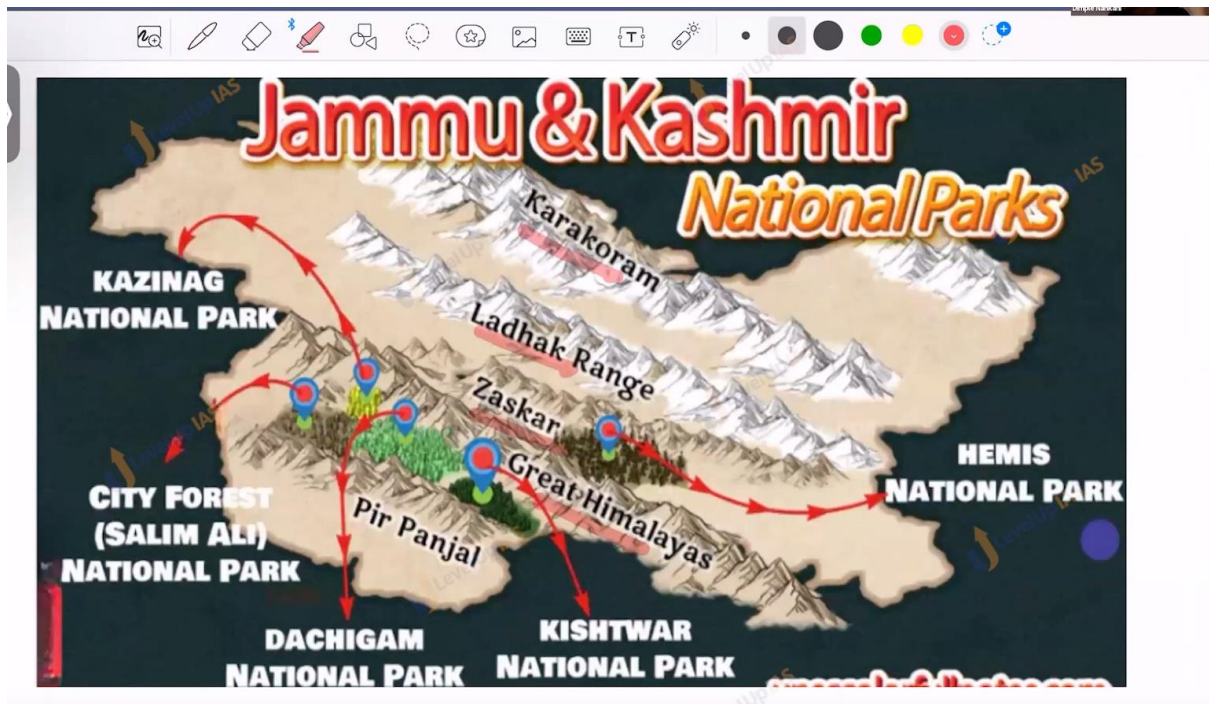


› Practise Question:

Q) This Flows entirely in India. This River meets Sutlej at Harike to form Harike wetland?

Q) This river has origin in Rakas Lake in Tibet and passes through Shipki la

Q) This river has Dal Lake on it and it forms border between India and Pakistan



BRAHMAPUTRA

- Origin: Chemayungdung glacier of Kailash range near Mansarovar lake.
- It travels eastward in Tibet (Known as Tsangpo)
- Enters Indian near Namcha Barwa
- The river is named of Siang or Dihang
- After meeting left bank tributaries of Dibang/Sikang and Lohit it is known as Brahmaputra.
- Major left bank tributaries: Burhi Dihing, Dhansari (South)
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2016: Brahmaputra River

Which of the following is/are tributary tributaries of Brahmaputra?

1. Dibang
2. Kameng
3. Lohit

Select the correct answer using the code given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

2014: Brahmaputra River

dnyaneshwar.mogal.ias@gmail.com(8421928891)

Consider the following rivers:

1. Barak
2. Lohit
3. Subansiri

Which of the above flows / flow through Arunachal Pradesh?

- (a) 1 only
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2014: Brahmaputra River

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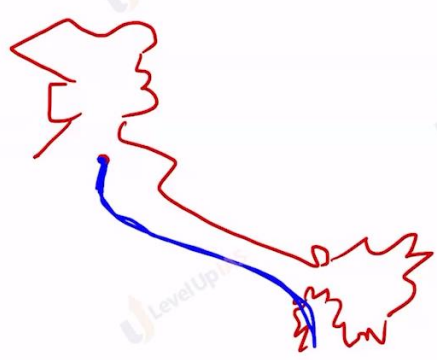
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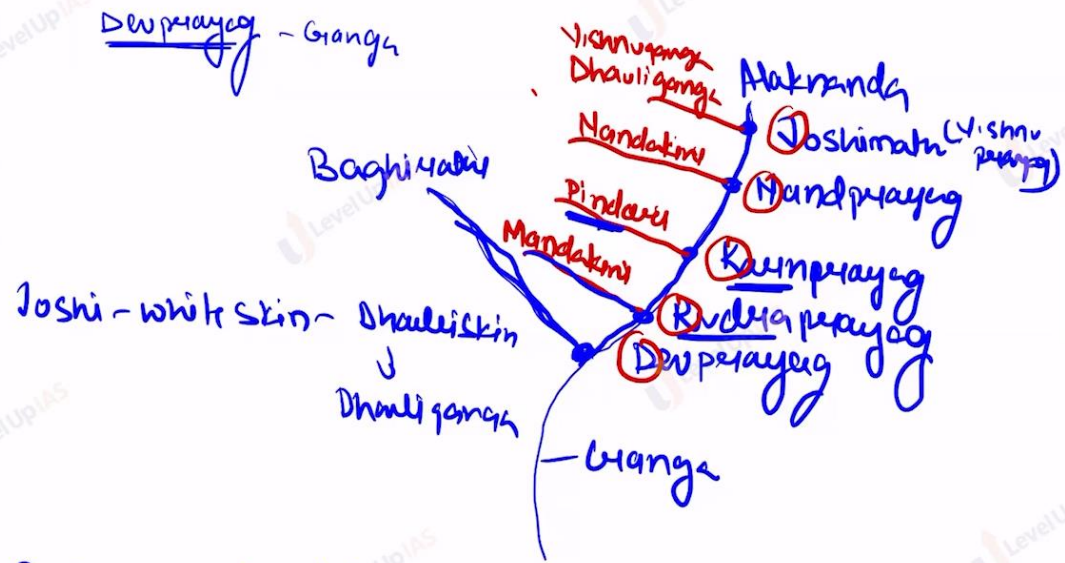
Indus ✓

- Beah ✓

- Ganga - origin - Gangotri
Grammohi



Devprayag - Ganga



Rudra - Mandakini
(Shiv) (Parvati)

Karna - Pindar (कर्ण पिन्डर)

Sources of Ganga:

- Bhagirathi Origin: Gangotri Glacier
- Alaknanda Origin: Saptopanth Glacier. It passes via Valley of Flowers
- **Left Bank Tributaries are:** Ramganga (origin: India), Gomti (origin: India), Ghagra (origin: Nepal), Gandhak (origin: Nepal), Kosi (origin: Nepal), Mahananda (origin: India)
- **Right Bank Tributaries:** Yamuna, Son

Ganga river:

- Origin: Gangotri Glacier near Gomukh in Uttarakhand where river Alaknanda meets Bhagirathi at Devprayag to form Ganga



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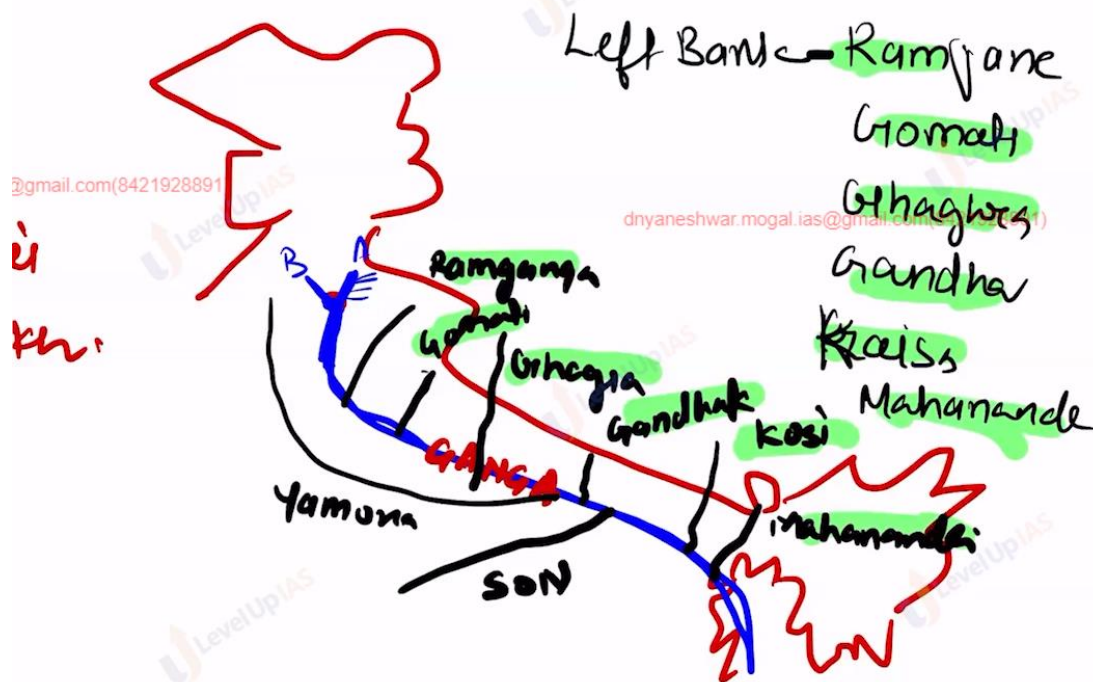
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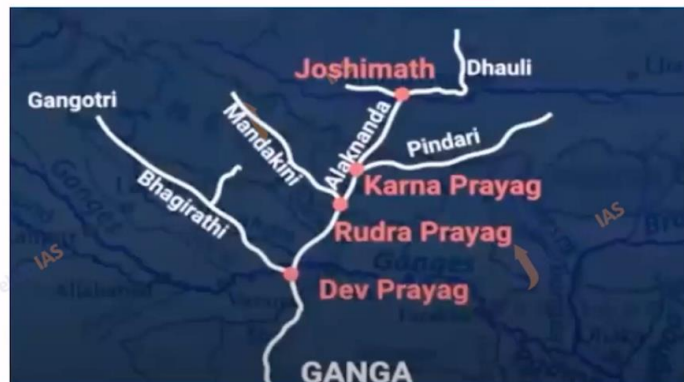




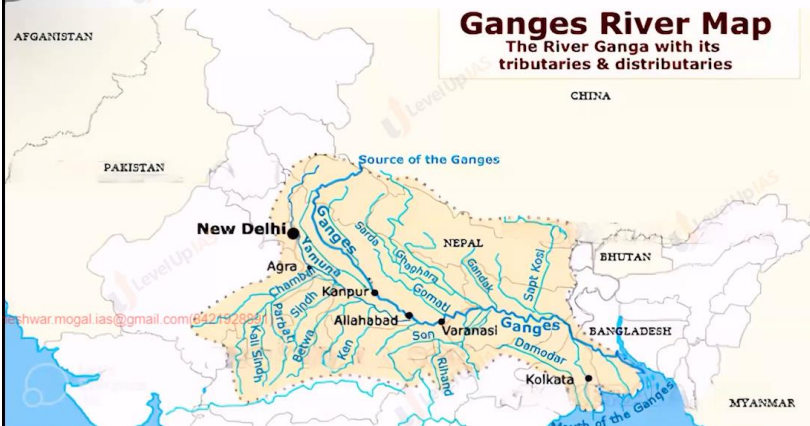
- **Important Lake:** ²⁰²³ Kanwar Lake. Asia's largest freshwater Oxbow Lake in India. Made by Gandhak River
- **Ganga Passes via Rajaji National Park (Utk)**
- **Ganga breaks at Malda Gap** into River Hooghly and River Padma
- **Sagar Island** is in Bay of Bengal near Ganga's Mouth

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GANGA RIVER SYSTEM

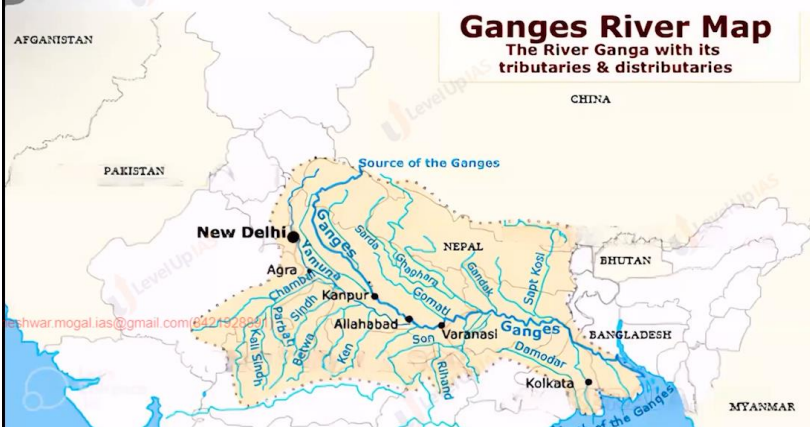


- **Important Dam:**
Tehri on River Bhagirathi (Highest Dam in India),
- **Narora Dam** on Ganga in U.P. Narora is also Nuclear Power Plant



- Barila wetland is at confluence of Ganga, Punpun and Gandhak in River;
- Uduwa Wetland which is the only bird sanctuary in Jharkhand

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Ramganga River:

- Origin: Namik Glacier near Garhwal Hills
- It passes via Corbet National Park

Gomti:

- Origin: Gomat Tal (U.P)

Ghagra:

- **Origin: Mapchachungo** Glacier near Mansarovar in Tibet
- By Volume, ghagra is largest tributary of Ganga

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Level Up Class

Ganges River Map

The River Ganga with its tributaries & distributaries

Gandhak:

- Origin: Nepal
- It meets Ganga at Patna
- It has Valmiki Nagar National Park/Tiger Reserve
- Lies between Dhaulagiri Peak and Mount Everest
- It divides Nepal into Half

#5) Kosi:

- Origin: Tibet
- Aka Sorrow of Bihar

#6) Mahananda:

- Origin: Darjeeling Hills

2013

Consider the following pairs

National Park	Rivers flowing through the Park
1. Corbett National Park	Ganga
2. Kaziranga National Park	Manas
3. Silent Valley National Park	Kaveri

Which of the above pair is/are correctly matched?

- 1 and 2
- 3 only
- 1 and 3
- None of the above

2013

Consider the following pairs

National Park	Rivers flowing through the Park
1. Corbett National Park	Ganga Ramganga
2. Kaziranga National Park	Manas X
3. Silent Valley National Park → Kerala	Kaveri - K & TN X

Which of the above pair is/are correctly matched?

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b) ~~3 only~~
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d) None of the above ✓

2023 Rivers

Consider the following statements:

1. Jhelum River passes through Wular lake. ✓
2. Krishna River directly feeds Kolleru Lake. ✗
3. Meandering of Gandak River formed Kanwar Lake. ✓

How many of the Statements given above are correct?

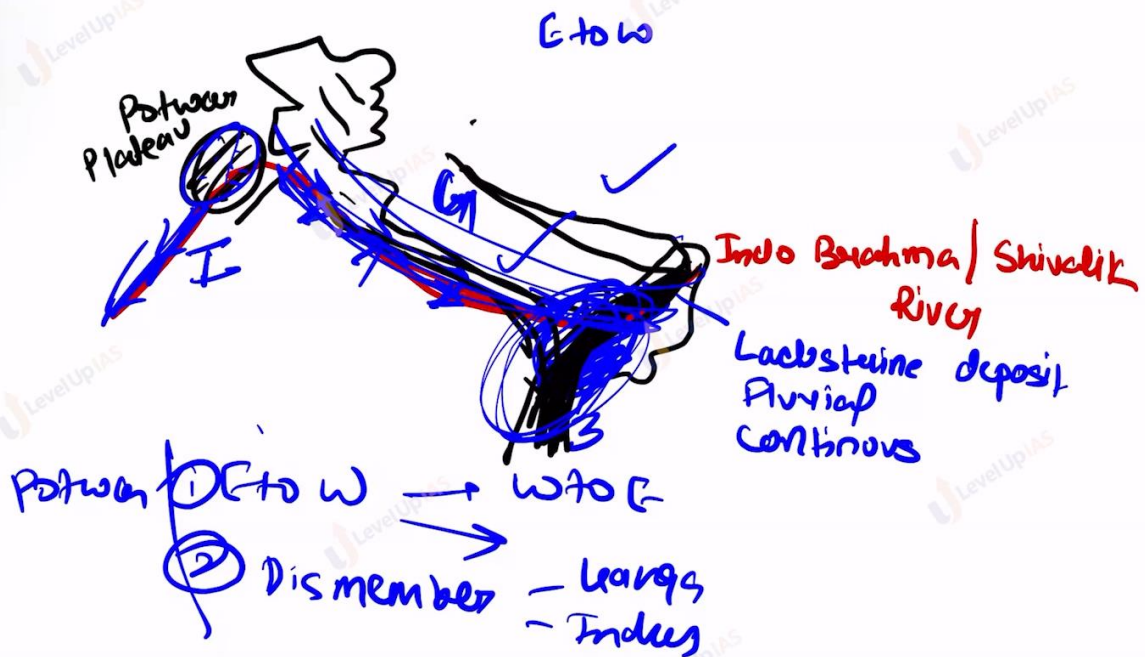
- a) ~~Only one~~
b) ~~Only two~~
c) All three
d) ~~None~~

Important Right Bank Tributary:

#1) Yamuna:

- Origin: Yamunotri Glacier from Banderpunch Peak in Uttarakhand
- Joins Ganga at Prayag (Allahabad) in U.P.
- Longest Tributary of Ganga
- Forms a border between UP and Haryana
- Passes via 4 states and 1 UT: Uttarakhand, Himachal Pradesh, Haryana, UP and Delhi
- Tributaries: **Left:** Chambal, Sindh, Betwa, Ken

Longest - Ghagra
Longest - Yamuna



Evolution of Himalayan Drainage:

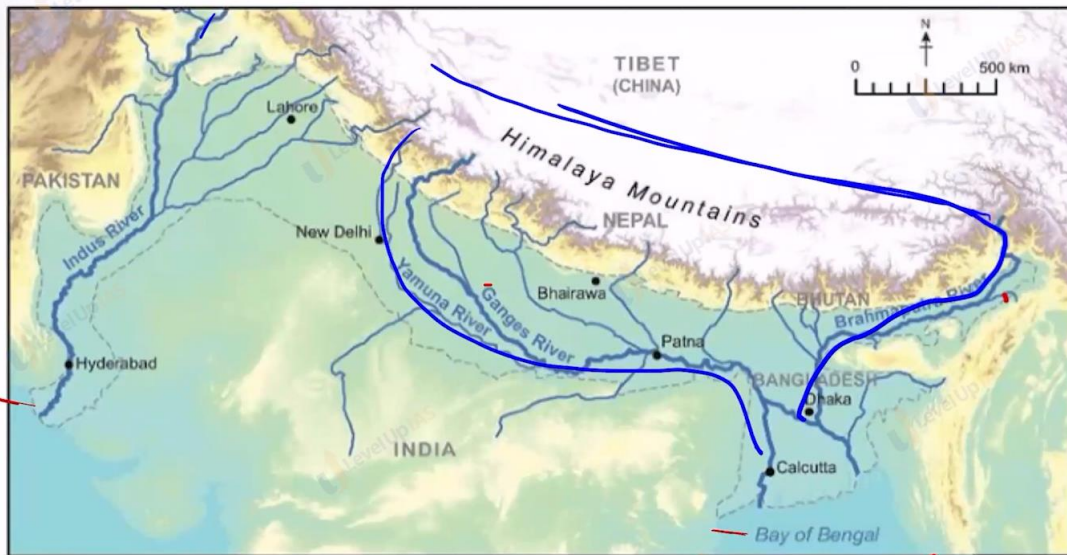
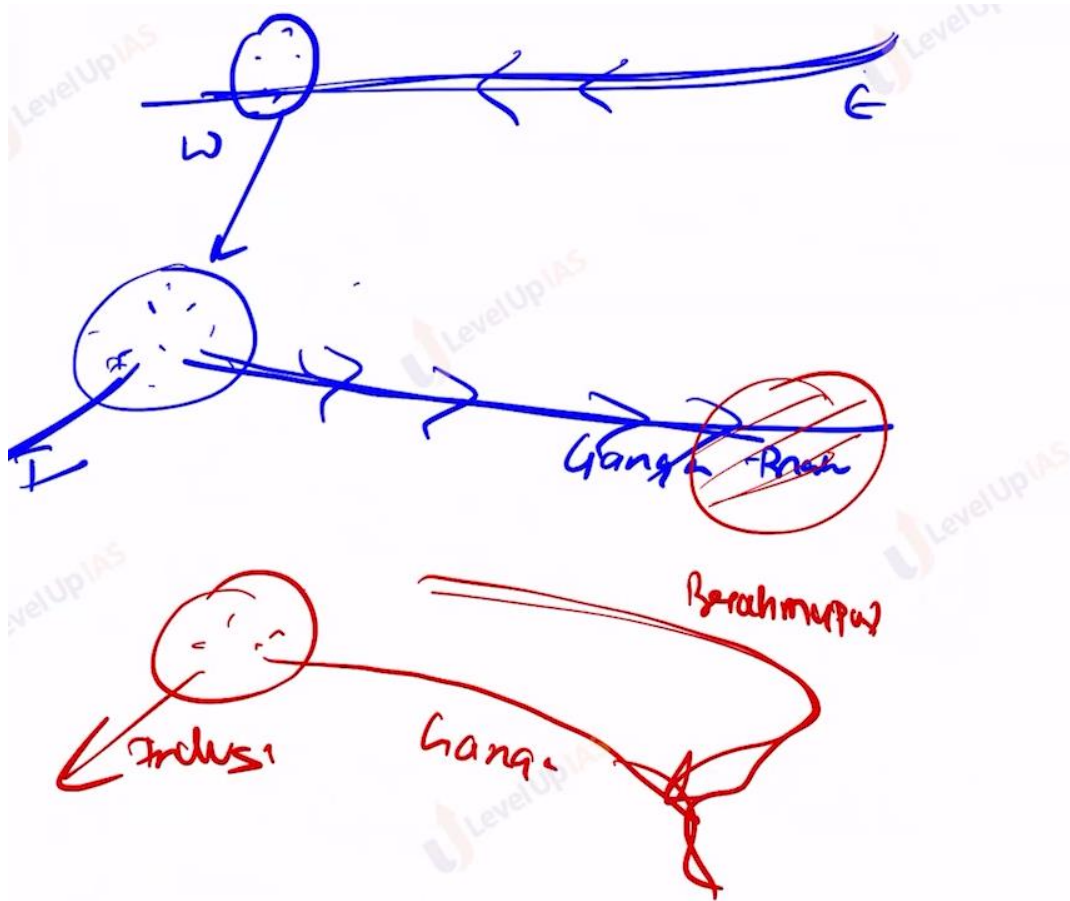
- Mighty River Shiwalik or Indo-Brahma traversed the entire longitudinal extent of the Himalaya from Assam to Punjab and Sindh, and finally discharged into the Gulf of Sind near lower Punjab during the Miocene period (5-24 million years ago)
- Continuity of Shiwalik and its lacustrine origin and alluvial deposits support this viewpoint.
- In time Indo– Brahma river was dismembered into three main drainage systems:
 - (i) The Indus and its five tributaries in the western part;
 - (ii) The Ganga and its Himalayan tributaries in the central part; and
 - (iii) Brahmaputra in Assam and its tributaries in the eastern part.
- The dismemberment was due to the Pleistocene upheaval of the Potwar Plateau (Delhi Ridge), which acted as the water divide between the Indus.
- Downthrusting of the Malda gap area between the Rajmahal hills and the Meghalaya plateau during the mid-pleistocene period, diverted the Ganga and the Brahmaputra systems to flow towards the Bay of Bengal

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Evolution of Peninsular Drainage:

- Three major geological events in the distant past have shaped the present drainage systems of Peninsular India:
- (i) Subsidence of the western flank of the Peninsula leading to its submergence below the sea during the early tertiary period. Generally, it has disturbed the symmetrical plan of the river on either side of the original watershed.
- (ii) Upheaval of the Himalayas when the northern flank of the Peninsular block was subjected to subsidence and the consequent trough faulting. The Narmada and The Tapi flow in trough faults and fill the original cracks with their detritus materials. Hence, there is a lack of alluvial and deltaic deposits in these rivers.
- (iii) Slight tilting of the Peninsular block from northwest to the southeastern direction gave orientation to the entire drainage system towards the Bay of Bengal during the same period.